

4 hours on examples + OTIS excerpt

Problem 1 (AIME 2019/14)

We want,

$$2019^8 + 1 \equiv 0 \pmod{p}$$

$$\Leftrightarrow 2019^{16} \equiv 1 \pmod{p}$$

Therefore, we need $\text{ord}_p(2019) = 16$. This means that $16 \mid p - 1$. Looking at the smallest two primes that satisfy this,

$$p = 17: 2019^8 + 1 \equiv 14 \pmod{17}$$

$$p = 97: 2019^8 + 1 \equiv 79^8 + 1 \equiv 0 \pmod{97}$$

Therefore, the answer is $p = 97$.

Problem 2 (Z762215A)

Note that multiples of p (i.e., $0 \pmod{p}$) cannot be primitive roots.

We find any g , a primitive root of p . We from Exercise 2.5 that, for all $k > 0$, g^k has order exactly $\frac{p-1}{\gcd(p-1, k)}$. On $0 \leq k < p - 1$, all g^k have distinct residues mod p , so they're $1, 2, \dots, p - 1 \pmod{p}$ in some order.

Therefore, we simply must count for how many k on $0 \leq k < p - 1$ where we have,

$$\frac{p-1}{\gcd(p-1, k)} = p-1$$

$$\Leftrightarrow \gcd(p-1, k) = 1$$

This count is hence simply $\phi(p-1)$.

Problem 3 (PUMaC 2016)

Define a function $f(a, b) = \begin{cases} 1, & \text{if } a^b \equiv 1 \pmod{2017} \\ 0, & \text{otherwise} \end{cases}$. Our desired quantity is,

$$\sum_{n=2018}^{2017^2} f(n, n)$$

Since, if $a = 2017k$, $f(a, b) = 0$, we can have,

$$= \sum_{a=1}^{2016} \sum_{b=1}^{2016} f(2017 \cdot a + b, 2017 \cdot a + b)$$

Using Fermat's Little Theorem, let $s = a + b$,

$$= \sum_{s=2}^{4032} \sum_{a=1}^{s-1} f(a, s)$$

Swapping the order of summations and taking mod 2016 by FLT,

$$= \sum_{a=1}^{2016} \sum_{b=0}^{2015} f(a, b)$$

$$= \sum_{a=1}^{2016} \frac{\phi(2017)}{\text{ord}_{2017}(a)}$$

Doing the same trick as last time,

$$= \sum_{a=1}^{2016} \frac{2016}{\left(\frac{2016}{\gcd(2016, a)}\right)}$$

$$= \sum_{a=1}^{2016} \gcd(2016, a)$$

Note that, $f(n) := \sum_{a=1}^n \gcd(n, a)$ is multiplicative,

$$= f(32) \cdot f(9) \cdot f(7)$$

$$f(32) = 1 + 2 + 1 + 4 + 1 + 2 + 1 + 8 + 1 + 2 + 1 + 4 + 1 + 2 + 1 + 16$$

$$+ 1 + 2 + 1 + 4 + 1 + 2 + 1 + 8 + 1 + 2 + 1 + 4 + 1 + 2 + 1 + 32 = 112$$

$$f(9) = 1 + 1 + 3 + 1 + 1 + 3 + 1 + 1 + 9 = 21$$

$$f(7) = 1 \cdot 6 + 7 = 13$$

The product is hence 30576.

10 clubs; 6 hours

Problem 5 (2012H01)

For the sake of contradiction, let there be, $n \mid 2^c + 3^d$, and we have $n \nmid 2^a 3^b + 1 \forall a, b \in \mathbb{Z}^{\geq 0}$.

We have,

$$2^c + 3^d \equiv 0 \pmod{n}$$

$$\Rightarrow 2^c 3^k + 3^{d+k} \equiv 0 \pmod{n}; \Rightarrow 2^{c+k} + 2^k 3^d \equiv 0 \pmod{n}$$

This means that we cannot find a $3^{d+k} \equiv 1 \pmod{n}$ or $2^{c+k} \equiv 1 \pmod{n}$. By Euler's Totient theorem, this would mean that $\gcd(3, n) > 1$ and $\gcd(2, n) > 1$, which means $6 \mid n$.

However, we claim that it's impossible to find $2^c + 3^d \equiv 0 \pmod{6}$.

If $c > 0; d > 0$, then $3^d \equiv 3 \pmod{6}$ and $2^c \equiv 2, 4 \pmod{6}$, so impossible.

If $c = 0$; then $2^0 + 3^d \equiv 1, 4 \pmod{6}$, so still impossible.

We hence have a contradiction, so we're done.

Note: For the following two problems, I have figured out the orders solution as well.

Problem 7 (Z2F0B95F)

For the sake of contradiction, let there exist a number g such that,

$g, g^2, g^3, \dots, g^{2^{n-1}}$ have remainder $1, \dots, 2^n - 1$ in some order.

This means that, $\text{ord}_{2^n}(g) = 2^{n-1}$. Obviously, $g^{2^{n-1}} \equiv 1 \pmod{2^n}$.

Note that $2^n - 1$ is odd. Since it has to appear, it must appear at $g^{2^{n-2}} \equiv -1 \pmod{2^n}$. If otherwise, then the order cannot be 2^{n-1} as $(2^n - 1)^2 \equiv 1 \pmod{2^n}$. Same is true for $2^{n-1} + 1$. The two numbers are distinct for $n \geq 3$ so we're done.

15 clubs; 7.5 hours

Problem 8 (Japan 1996/2)

Clearly, neither are divisible by 5 or 7.

For any other prime p ,

$$5^m + 7^m \equiv 0 \pmod{p}$$

$$\Rightarrow 5^{2m} \equiv 7^{2m} \pmod{p}$$

Similarly, $7^{2n} \equiv 5^{2n} \pmod{p}$. Therefore, since both are invertible, $\forall k \in \mathbb{Z}$,

$$7^{2n+2km} \equiv 5^{2n+2km} \pmod{p}$$

We can use the Euclidean Algorithm to get,

$$7^{2 \gcd(m,n)} \equiv 5^{2 \gcd(m,n)} \pmod{p}$$

$$\Leftrightarrow 49 \equiv 25 \pmod{p}$$

Therefore, this is impossible for $p > 7$.

We can easily check that it's impossible for the gcd to be a multiple of 8 and 9, but both are definitely divisible by 12, so we're done.

Problem 9 (SPBRG)

Since $\gcd(a, a^n - 1) = \gcd(a, 1) = 1$,

$$a^{\phi(a^n - 1)} \equiv 1 \pmod{a^n - 1}$$

In addition, obviously, $\text{ord}_{a^n - 1}(a) = n$ (n obviously works, and any lower is less than $a^n - 1$).

Therefore, $n \mid \phi(a^n - 1)$.

21 clubs; 8.5 hours

Problem 12 (IMO 2005/4)

We claim the only solution is 1, which obviously works.

We prove that, for every prime p , there exists an a_n where $p \mid a_n$.

For $p = 2, 3$, we can just use a_2 .

For any other p , we claim that,

$$2^{p-2} + 3^{p-2} + 6^{p-2} - 1 \equiv 0 \pmod{p}$$

Note that a^{p-2} for $\gcd(a, p) = 1$ is the unique modular inverse of $a \pmod{p}$.

If $p \equiv 1 \pmod{6}$, we write $p = 6k + 1$ to get,

$$\equiv (3k + 1) + (4k + 1) + (5k + 1) - 1 \equiv 12k + 2 \equiv 0 \pmod{6k + 1}$$

For the other case, $p \equiv 5 \pmod{6}$, we write $p = 6k + 5$ to get,

$$\equiv (3k + 3) + (2k + 2) + (k + 1) - 1 \equiv 0 \pmod{6k + 5}$$

In either case, we establish divisibility.

Problem 14 (11MOPR34)

$$\text{ord}_p(2) \mid n$$

$$\Rightarrow 2^{\text{ord}_p(2)} - 1 \mid 2^n - 1$$

$$\Rightarrow v_p(2^{\text{ord}_p(2)} - 1) = 1$$

Note that,

$$2^{p-1} - 1 = (2^{\text{ord}_p(2)} - 1)(1 + 2^{\text{ord}_p(2)} + 2^{2 \cdot \text{ord}_p(2)} + \dots + 2^{p-1})$$

However, $1 + 2^{\text{ord}_p(2)} + 2^{2 \cdot \text{ord}_p(2)} + \dots + 2^{p-1}$ is divisible by p if and only if $\text{ord}_p(2) = 1$,

which is impossible. We're therefore done.

29 clubs; 9.5 hours

Problem 16 (Putnam 2021 A5)

By CRT, we can independently check for divisibility by 43 and 47.

$$\begin{aligned}\sum_{n \in A} n^j &\equiv 47 \cdot (1^j + 2^j + \dots + 42^j) - (47^j + 94^j + \dots + 1974^j) \\ &\equiv 46(1^j + 2^j + \dots + 42^j) \equiv 1^j + 2^j + \dots + 42^j \equiv 0 \pmod{43}\end{aligned}$$

Similarly,

$$\sum_{n \in A} n^j \equiv 1^j + 2^j + \dots + 46^j \equiv 0 \pmod{47}$$

We claim the answer is all j where $42 \nmid j$; $46 \nmid j$. Obviously, by FLT, if either $42 \mid j$ or $46 \mid j$, then it's impossible.

We now show all other j do work. The numbers $\{1, 2, \dots, 42\} = \{k^0, k^1, \dots, k^{41}\}$ for some primitive root k of 43. Therefore,

$$\sum_{n \in A} n^j \equiv k^0 + k^j + k^{2j} + \dots + k^{41j} \equiv \frac{k^{42} - 1}{k^j - 1} \pmod{43}$$

If $42 \nmid j$, then $k^j - 1 \not\equiv 0 \pmod{43}$ has an inverse, and $k^{42j} - 1 \equiv 0 \pmod{43}$, so we're good.

Similarly, if $46 \nmid j$, $\sum_{n \in A} n^j \equiv 0 \pmod{47}$.

Problem 18 (HMMT November 2014)

For all odd m , we plug in $n = m - 1$ to get, $n^{n+1} + 1 \equiv (-1)^m + 1 \equiv 0 \pmod{m}$.

Note that m cannot be a multiple of 4 since we can casework to get $n^{n+1} + 1 \equiv 0 \pmod{4}$ is impossible.

For $m = 2k$ for some positive integer k , if there exists an n , then n must be odd since otherwise $n^{n+1} + 1$ is odd. Therefore, $n + 1$ is even.

By Fermat's Christmas Theorem, we hence get it's impossible for any prime $p \equiv 3 \pmod{4}$ to divide $n^{n+1} + 1$, so it's impossible for any prime $p \equiv 3 \pmod{4}$ to divide m .

Apart from the 25 odd m in the range, we are left with,

$$m = 2, 10, 26, 38, 50$$

We can that each work.

For $m = 2$, $n = 1$. For $m = 10$, $n = 17$ works. For $m = 26$, $n = 5$. For $m = 38$, $n = 7$. For $m = 50$, $n = 57$.

37 clubs; 11 hours

Problem 20 (ZEF6368F)

Claim 1: $v_p(n) \leq 1 \forall p$.

If $v_p(n) \geq 2$ for some p , we get, for all a and $k = v_p(n) \geq 2$.

$$a^{n+1} \equiv a \pmod{p^k}$$

However, plugging in $a = p^{k-1}$, since $n + 1 \geq 2$, contradiction.

Therefore, we can write $n = p_1 p_2 \dots p_k$ for distinct primes $p_1, \dots, p_k$.

Claim 2: $p_i - 1 \mid n \forall p_i$

Taking mod p_i ,

$$a^{n+1} \equiv a \pmod{p_i}$$

If $a \equiv 0 \pmod{p_i}$, we're done. If not, then a is invertible, so,

$$a^n \equiv 1 \pmod{p_i}$$

Since there exists a primitive root mod p , we plug in that primitive root to get,

$$p_i - 1 \mid n$$

If both claims are met, then n is valid by CRT. We claim the answer is,

$$S = \{2, 6, 42, 1806\}$$

We can check that all of these are multiples of one less than each of their prime factors.

Claim 3: If we already have

To prove uniqueness, we now construct all possible n . Since $n \geq 2$, it has at least one prime factor. We see that $n = 2$ is our first solution.

If n has more than one prime factor, then let p be the next biggest prime factor. Since $p - 1$ cannot divide itself nor any greater prime factors, $p - 1 \mid 2$, so only $p = 3$ can be the next biggest prime factor. Immediately, we find that $n = 6$ works.

If n has more than two prime factors, then we show that $7 \mid n$ as well. We have $p - 1 \mid 6$. We see that only $p = 7$ can be the next biggest prime factor. Immediately, we find $n = 42$ also works.

Continuing, we then get $p - 1 \mid 42$ and find next biggest prime factor can only be $p = 43$. We find that $n = 1806$ is a solution.

We finally get that $p - 1 \mid 1806$. We can check that there exist no greater prime divisors which work.

Problem 22 (USA TST 2008/4)

We can easily check that non-positive n don't work since $n < -1$ makes RHS negative; $n = -1$ and $n = 0$ don't work.

For the sake of contradiction, let there exist integers a, n such that,

$$\begin{aligned} a^2 &= n^7 + 7 \\ \Leftrightarrow a^2 + 11^2 &= n^7 + 2^7 \end{aligned}$$

Taking mod 4, we see that $n \equiv 1 \pmod{4}$ since the LHS cannot be 3 mod 4.

Note that we can factor,

$$n^7 + 2^7 = (n + 2)(n^6 + 2n^5 + 4n^4 + \dots + 2^6)$$

Since $n + 2 \equiv 3 \pmod{4}$, there exists a prime $p \mid n + 2$ where $p \equiv 3 \pmod{4}$.

However, by Fermat's Christmas Theorem, unless if $p \mid \gcd(a, 11)$, it is impossible for,

$$p \mid a^2 + 11^2 \Leftarrow p \mid n + 2$$

Therefore, given $p \mid \gcd(a, 11)$, we get $p = 11$ is the only prime that is 3 mod 4 dividing both sides. We have, for $k := \frac{a}{11}$, since -1 is not a quadratic residue mod 11,

$$v_{11}(a^2 + 11^2) = v_{11}(11^2 \cdot (1 + k^2)) = 2$$

However, since $n + 2 \equiv 3 \pmod{4}$, and 11 is the only prime that is 3 mod 4 dividing both sides, $v_{11}(n + 2)$ must be odd. Therefore, by LTE,

$$2 = v_{11}(a^2 + 11^2) = v_{11}(n^7 + 2^7) = v_{11}(n + 2) + v_{11}(7) = v_{11}(n + 2) \equiv 1 \pmod{2} (!)$$

Problem 23 (Krishna Pothapragada)

For the sake of contradiction, let $d(n) \mid n^2 + 1$.

Lemma: Given p an odd prime, a, b not divisible by p , and a positive integer k such that we have, $a^{2^k} + b^{2^k} \equiv 0 \pmod{p}$, then $v_2(p - 1) \geq k + 1$.

Proof: Since b is invertible, we let $x = ab^{-1} \pmod{p}$,

$$\begin{aligned} a^{2^k} + b^{2^k} &\equiv 0 \pmod{p} \Leftrightarrow x^{2^k} + 1 \equiv 0 \pmod{p} \\ &\Leftrightarrow x^{2^k} \equiv -1 \pmod{p} \\ &\Rightarrow x^{2^{k+1}} \equiv 1 \pmod{p} \\ &\therefore \text{ord}_p(x) = 2^{k+1} \\ &\Leftrightarrow 2^{k+1} \mid p-1 \Leftrightarrow v_2(p-1) \geq k+1 \end{aligned}$$

We now use induction to prove that, for any prime divisors of p of $d(n)$, $v_2(p-1) = k$ and $n = a^{2^{k-1}}$ for some integer a is satisfied for all k .

Base case: $k = 2$

Since n is even, $n^2 + 1$ can only have prime divisors which are $\equiv 1 \pmod{4}$ by our lemma, so $v_2(p-1) = 2$ is satisfied.

In addition, since, if n is not a square, then $d(n)$ is even and cannot divide $n^2 + 1$, which is odd. Therefore, $n = a^2$ for some a .

Inductive step: $v_2(p-1) = k$ to $v_2(p-1) = k+1$

We write the prime factorization of n ,

$$n = p_1^{i_1} p_2^{i_2} \dots p_m^{i_m}$$

Since $n = a^{2^{k-1}}$ for some integer a , we can write $i_x = 2^{k-1} j_x$ for all integers $1 \leq x \leq m$.

$$n = p_1^{2^{k-1} j_1} p_2^{2^{k-1} j_2} \dots p_m^{2^{k-1} j_m}$$

We know that,

$$\begin{aligned} d(n) \mid p_1^{2^k j_1} p_2^{2^k j_2} \dots p_m^{2^k j_m} + 1^{2^k} \\ \Leftrightarrow (2^{k-1} j_1 + 1)(2^{k-1} j_2 + 1) \dots (2^{k-1} j_m + 1) \mid p_1^{2^k j_1} p_2^{2^k j_2} \dots p_m^{2^k j_m} + 1^{2^k} \end{aligned}$$

By our lemma, any prime divisors of $p_1^{2^k j_1} p_2^{2^k j_2} \dots p_m^{2^k j_m} + 1^{2^k}$ must be, $p \equiv 1 \pmod{2^{k+1}}$.

Hence, if any of the j_1 are odd, we'd have a contradiction. Therefore, we get that we can write,

$n = a^{2^k}$ for some a .

60 clubs; 15 hours